

SurakshaNet: on-device Hinglish abuse detection

B.Tech mid-semester briefing

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CHROME MV3 INFERENCE



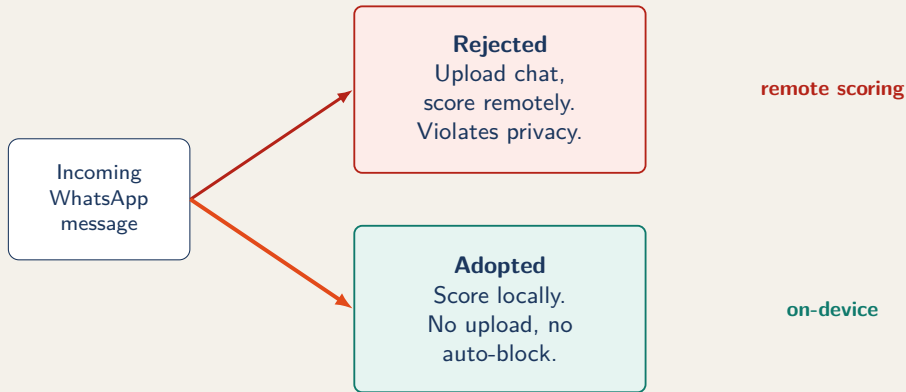
Conversion is training-time. Inference never calls Qwen.

Overview



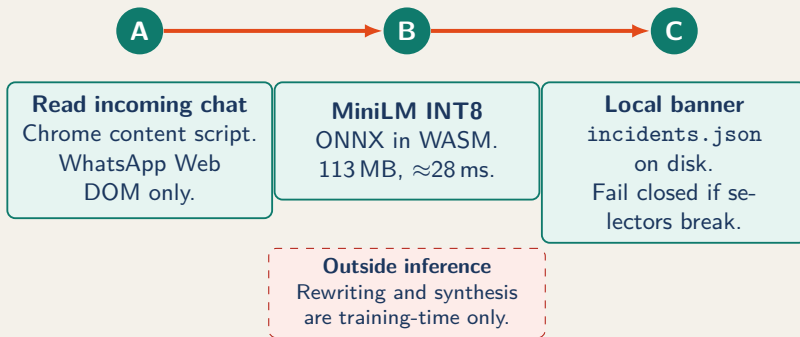
Inference remains the shipped MiniLM path. The Hinglish contribution is a training-set construction that does not alter A–B–C.

On-device scoring; a cloud round-trip is not a valid design.



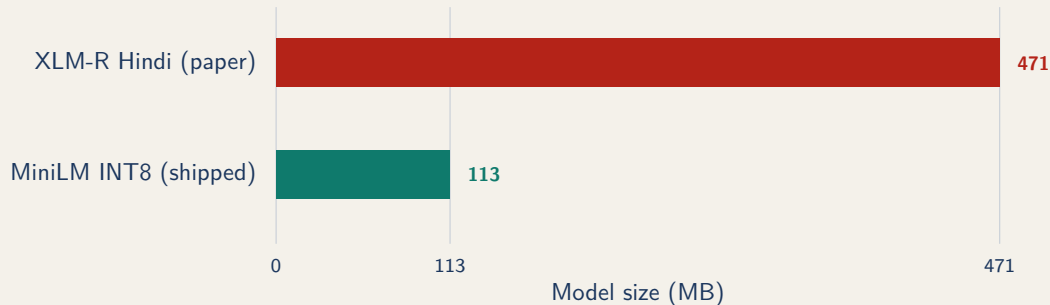
Term — on-device. *The model, the score, and incidents.json remain on the user's laptop.*

Inference path A–B–C; conversion is training-time only.



Term — Chrome MV3. *Current extension platform. Caps bundle size and forbids a silent cloud round-trip at inference.*

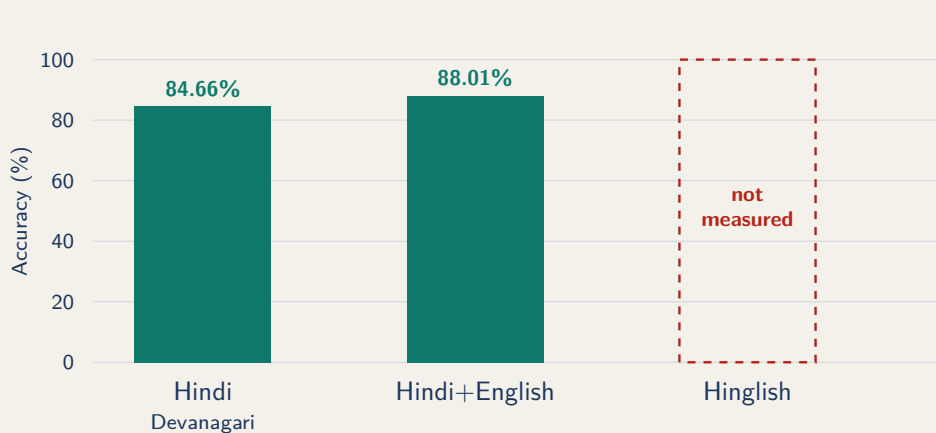
XLM-R exceeds the Chrome size budget.



MACD XLM-R Hindi ceiling is 86.34%. MiniLM INT8: poster Hindi 84.66%, drop under 0.11 pp, 28 ms.

Term — INT8. 8-bit quantized ONNX that Chrome loads. FP32 is the training checkpoint.

Poster evaluation omitted Hinglish.



The dashed bar is a missing measurement, not a score. Subsequent slides fill that cell without changing A-B-C.

Labeled MACD is Devanagari; traffic is Roman Hinglish.



Term — Hinglish. *Hindi as the matrix language, written in Roman letters, with English words mixed in. Not ITRANS, and not Devanagari in another font.*

Training uses published MACD and Davidson labels.



Term — MACD. *Multilingual Abusive Comment Detection. ShareChat-scale Hindi comments with published labels.*

Research question

Can MACD Hindi be rewritten as conversational Hinglish with published labels preserved, then measured on a test frozen before training?

Constraint 1

Inference path A-B-C is unchanged.

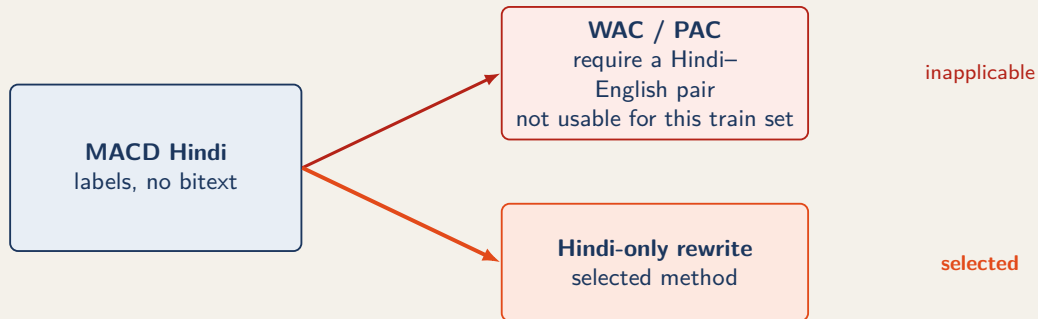
Constraint 2

Class 0 remains abusive after rewrite.

Constraint 3

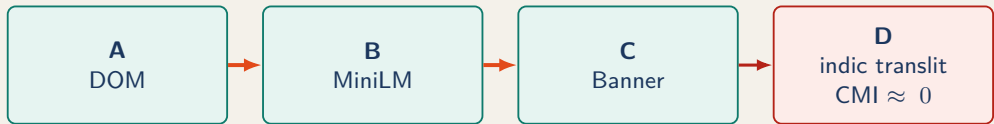
Hinglish test $n = 6368$ is frozen first.

WAC and PAC require bitext; MACD provides Hindi only.



Term — HinGE / WAC / PAC. *HinGE: human Hinglish benchmark, $n \approx 1964$ (IITGN). WAC: word-aligned mixing. PAC: phrase-aligned mixing. Both require bitext. HinGE is used later as converter evaluation, not as a train-set builder.*

Script transliteration is not code-mixing.



rejected as a training method

Hypothesis. Cheap, no GPU. If the language mixed, CMI would rise and WER against ITRANS would stay low.

Result. The sentence is still Hindi in another script. Script conversion $\neq$ code-mixing.

Term — CMI. *Code-mixing index (Das & Gambäck). 0 = one language. Higher = deeper mix.*

Term — WER. *Word error rate $(S + D + I)/N$. Low WER vs ITRANS means “just transliteration.”*

Only conversational Hinglish matches the target register.

1. MACD Hindi

कल मीटिंग है चार बजे

Labeled corpus.
Devanagari. Gold class.

2. indic_transliteration

kala mITiMga
hai cAra baje

Rejected. Same
Hindi. CMI $\approx$ 0.

3. Conversational mix

kal meeting hai
4 baje

Target. English words,
Hindi grammar.

Qwen3-32B rewrite is offline; it is not in Chrome inference.

INFERENCE — Qwen is not called

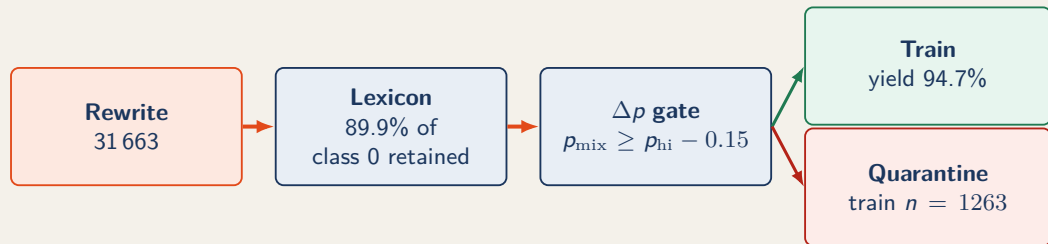


TRAINING — Bedrock Mantle, us-east-1, once



31 663 lines converted. Qwen is a converter, not a detector, and it is not inside Chrome.

Quarantine preserves class-0 labels after sanitisation.



Term — quarantine. *Held-out rejected rewrites with a reason: score drop, missing lexicon stem, collapsed length, refusal, leftover Devanagari. Not deleted. Not in train.*

Hinglish evaluation is frozen at $n = 6368$ before training.

Protocol

Train cleaning is allowed.
Quarantine is logged, not deleted.
Synthetic rows stay out of val/test.
CMI is reported even when low.

Forbidden

Rewriting the frozen 6 368 after error analysis.
Shipping a nearby ablation.
Placing Qwen inside Chrome.

Moving the test after error analysis would make 84.36% incomparable to the poster. Layer-1 cleaning and 977 synthetic rows were tried on train only.

Omitting Hindi MACD collapses Devanagari accuracy to 70.04%.

Failed mix

Train = Davidson 80% + Hinglish only.
Hindi test 70.04% ($n = 6728$).
Poster Hindi accuracy collapsed.

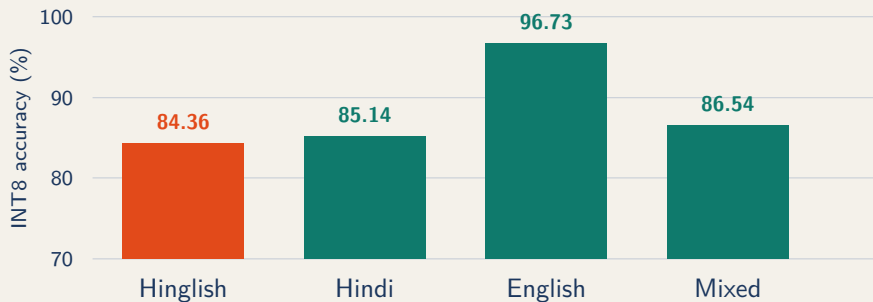


Corrected protocol

Train = Hindi MACD +
Davidson 80% + gated Hinglish.
Hindi INT8 85.14%. Shipped mix.

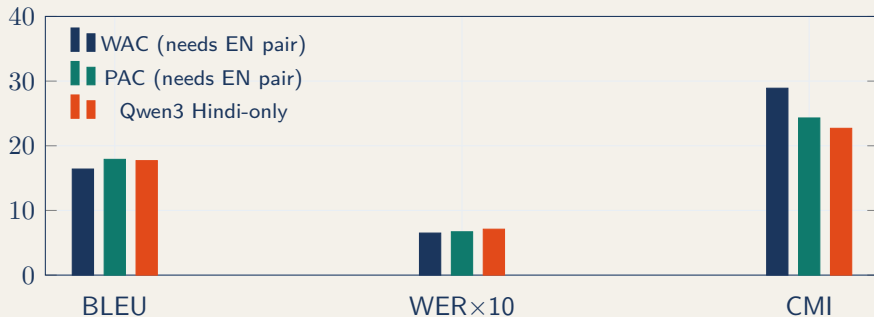
Corrected sizes: train 58 929 / val 15 581 / test 15 575. A Hinglish gain that costs Hindi is not adopted.

Shipped INT8 Hinglish accuracy is 84.36% with A–B–C unchanged.



Frozen Hinglish $n = 6368$. Mixed 86.54% is lower than poster 88.01% because Hinglish is now inside the mix. Same MiniLM, 113 MB, 28 ms.

HinGE: Hindi-only Qwen versus WAC and PAC

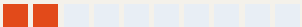


Human HinGE CMI = 35.6. This table evaluates the *converter*. It does not replace detector accuracy. **Term** — **BLEU**. Precision against a human Hinglish reference. Not abuse-detection accuracy.

MACD rewrites mix more shallowly than HinGE human references.

$$\text{CMI} = 100 \left(1 - \frac{\max_i w_i}{n} \right)$$

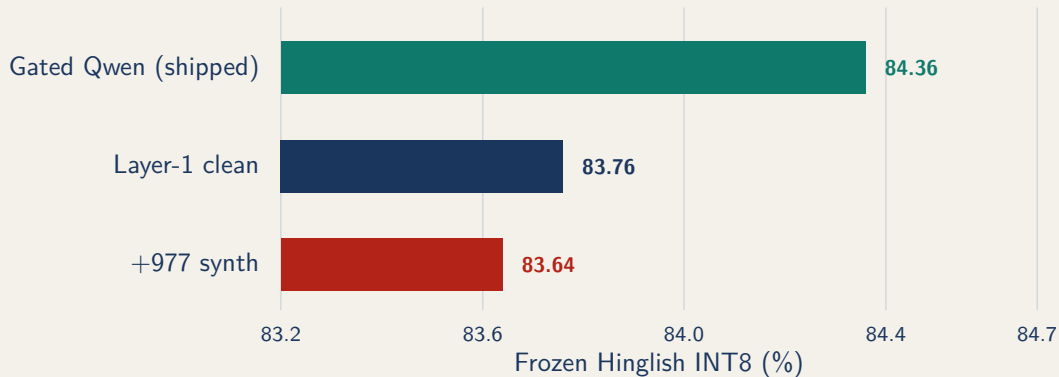
n : language-tagged tokens (no emojis). w_i : count in language i . CMI = 0 if only one language remains.

Qwen MACD  13.7

Humans  35.6

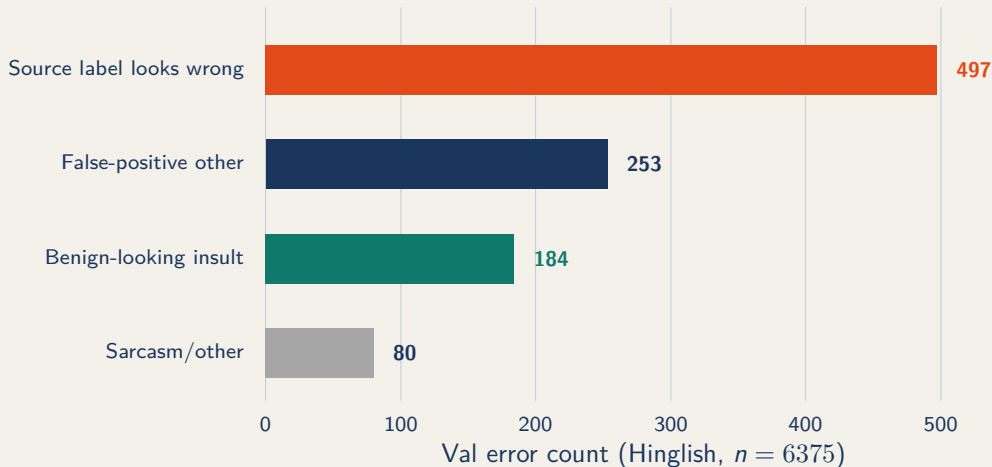
25% of Qwen MACD lines have CMI = 0 (romanized Hindi or English, not a mix).

Layer-1 and synthetic rows do not improve the freeze.



Deltas -0.60 and -0.72 on the same freeze. Synthetic rows never entered test. The nearby recipes are not shipped.

Largest validation errors are noisy MACD gold, not obfuscation.



507 false positives, 528 missed abuse. Cluster labels are a research reading, not a second gold. Cleaning this on the *test* set is exactly what the freeze forbids.

Closed results and remaining measurement

Closed

On-device A-B-C, 113 MB, 28 ms.

Hinglish 84.36% on a frozen test.

Hindi recovered after 70.04%.

Gated Qwen; transliteration and synth rejected.

Open

Raise CMI toward HinGE humans (35.6)
without sanitising class 0,
without unfreezing $n = 6368$,
and without placing Qwen in Chrome.